

Dürtü katarı

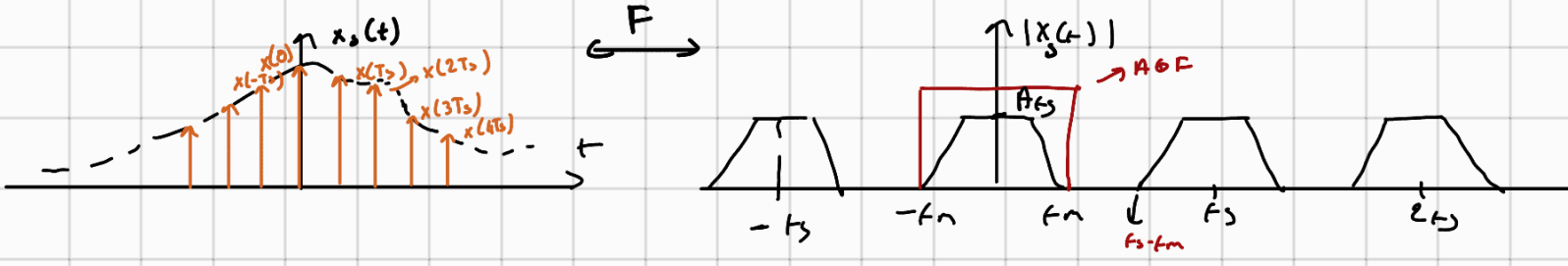
$$x_\delta(t) = \sum_{n=-\infty}^{+\infty} \delta(t-nT_s) \quad \overset{F}{\longleftrightarrow} \quad X_\delta(t) = \frac{1}{T_s} \sum_{n=-\infty}^{+\infty} \delta(t-nf_s) \quad (= f_s)$$



$$x_s(t) = x(t) \cdot x_\delta(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t-nT_s) = \sum_{n=-\infty}^{+\infty} x(nT_s)\,\delta(t-nT_s)$$

↓ Örneklenmiş işaret (Dürtü katarı)

$(f(t)\,\delta(t-t_0) = f(t_0)\,\delta(t-t_0))$



$f_s - f_m \geq f_m$

$f_s \geq 2f_m$

$f_s \geq (f_N = 2f_m)$

$$x_s(t) = x(t) \cdot x_\delta(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t-nT_s)$$

$$= \sum_{n=-\infty}^{\infty} x(nT_s)\,\delta(t-nT_s)$$

$$f(t)\,\delta(t-t_0) = f(t_0)\,\delta(t-t_0)$$

↕ F

$$X_s(f) = X(f) \circledast X_\delta(f)$$

$$= X(f) \circledast f_s \sum_{n=-\infty}^{\infty} \delta(f-nF_s)$$

$$= f_s \sum_{n=-\infty}^{+\infty} X(f) \circledast \delta(f-nf_s)$$

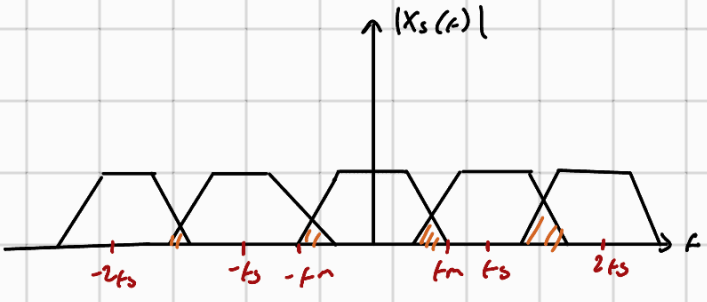
$$X_s(f) = f_s \sum_{n=-\infty}^{\infty} X(f-nf_s)$$

$$X(f) \circledast \delta(f-f_0) = X(f-f_0)$$

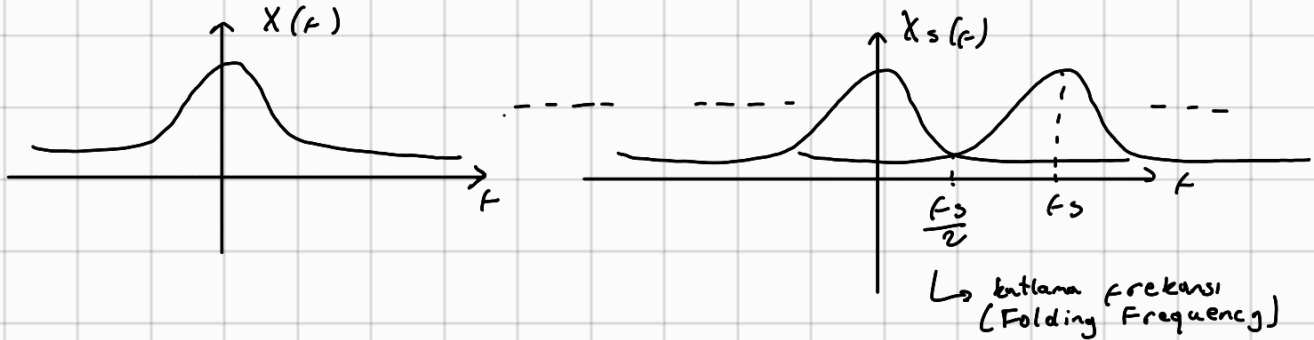
sbt

$f_s < 2f_m$ olursa

Örtüşme (Aliasing)

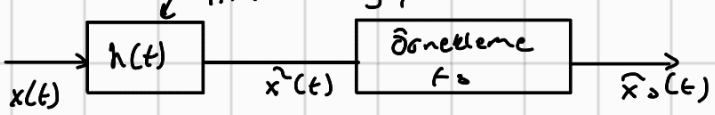


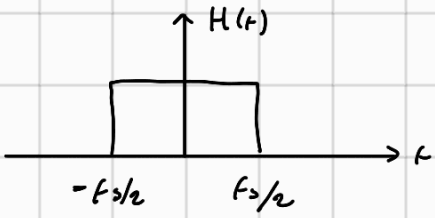
Pratikte $x(t)$ zamanda sınırlı frekansta sınırlı değil (not band limited)



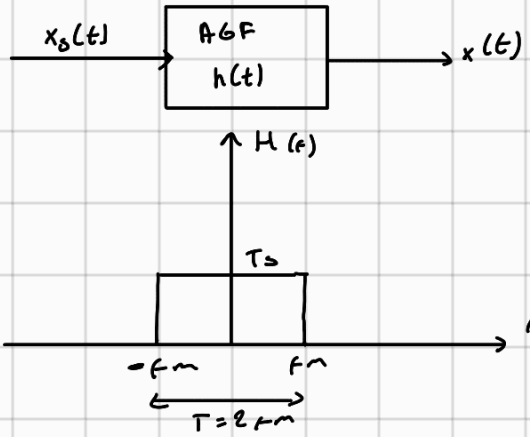
↳ katlama frekansı (Folding Frequency)

Çözüm: örtüşme engelleyici filtre
↙ Anti-aliasing filter





İşareti yeniden oluşturma
(Signal reconstruction)



$$H(f) = A\,\Pi\left(\frac{f}{T}\right) = T_s\,\Pi\left(\frac{f}{2f_m}\right)$$

$\downarrow F^{-1}$

$$h(t) = 2f_m T_s \operatorname{sinc}(2f_m t)$$

Eğer $f_s = 2f_m \rightarrow 1$ olur

$$A\,\Pi\left(\frac{t}{T}\right) \overset{F}{\leftrightarrow} AT\,\operatorname{sinc}(fT)$$
$$AT\,\operatorname{sinc}(tT) \overset{F}{\leftrightarrow} A\,\Pi\left(\frac{f}{T}\right)$$

$$x(t) = x_s(t) \circledast h(t)$$

$$= h(t) \circledast x_s(t) = h(t) \circledast \sum_{n=-\infty}^{\infty} x(nT_s)\,\delta(t-nT_s)$$

$$= \sum_{n=-\infty}^{\infty} x(nT_s)\,[h(t) \circledast \delta(t-nT_s)]$$

$h(t-nT_s)$

üçgen ise linear interpolasyon

Π ise kare darbelerle temsil edilir

(geriden düşürme denklemi) Reconstruction eq

$$x(t) = \sum_{-\infty}^{\infty} x(nT_s)\,h(t-nT_s) = 2f_m T_s \sum_{-\infty} x(nT_s)\,\operatorname{sinc}[2f_m(t-nT_s)]$$

(İdeal interpolasyon ya da ideal interpolation)



Örnek: Lathi, Prob. 6.1-4

A signal $x(t) = \text{sinc}^2(5t)$ is sampled (using uniformly spaced impulses) at a rate of

i) 5 Hz, ii) 10 Hz, iii) 20 Hz

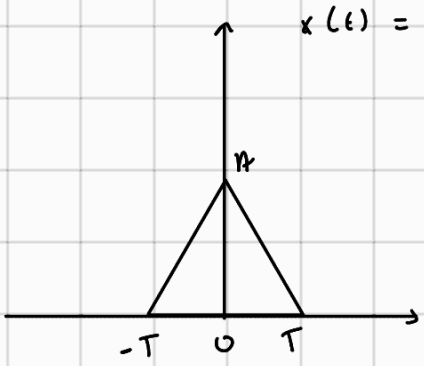
For each of three cases

a) Sketch the sampled signal

b) Sketch the spectrum of the sampled signal

c) Explain whether you can recover the signal $x(t)$ from the sampled signal

d) If the sampled signal is passed through an ideal LPF of BW 5 Hz, Sketch the spectrum of the output signal.



$x(t) = A\,\Lambda\left(\frac{t}{T}\right)$

$\longrightarrow X(f) = AT\,\text{sinc}^2(fT)$

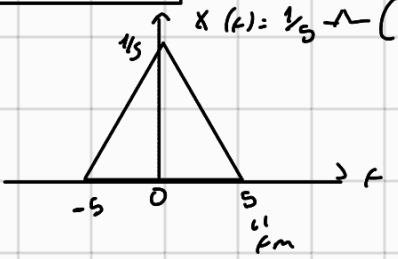
$$AT\,\text{sinc}^2(tT) \overset{F}{\longleftrightarrow} A\,\Lambda\left(\frac{f}{T}\right)$$

$x(t) = \text{sinc}^2\left(\frac{5}{T}t\right)$

$A = \frac{1}{1} = \frac{1}{5}$

$\overset{F}{\longleftrightarrow}$

$X(f) = \frac{1}{5}\,\Lambda\left(\frac{f}{5}\right)$



(i) $f_s = 5$ Hz, $T_s = \frac{1}{f_s} = \frac{1}{5}$ sn

$x_\delta(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$ → 1/5

$x_s(t) = x(t)\,x_\delta(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\,\delta(t - nT_s)$

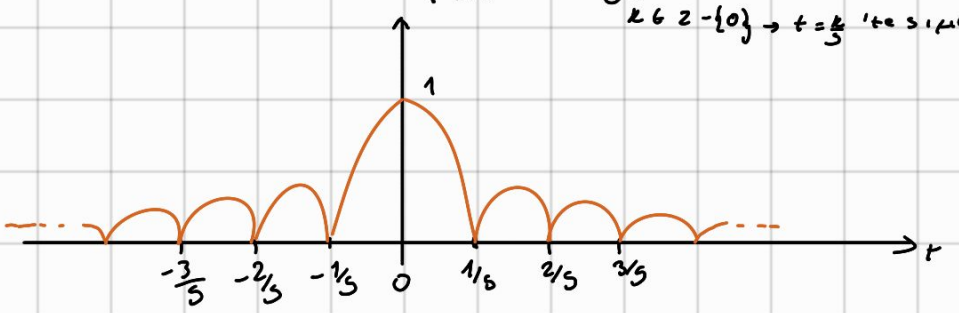
$X_s(f) = f_s \sum_{n=-\infty}^{\infty} X\left(f - n\frac{f_s}{5}\right)$

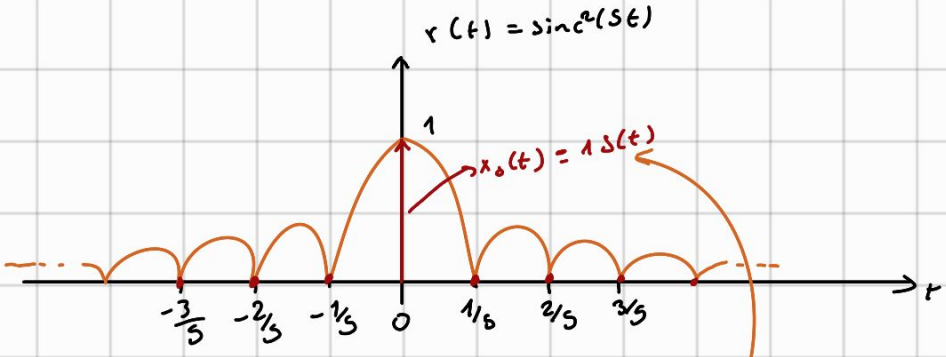
a)

$r(t) = \text{sinc}^2(5t)$ 0 geçişler $\frac{1}{5}$ katlarında

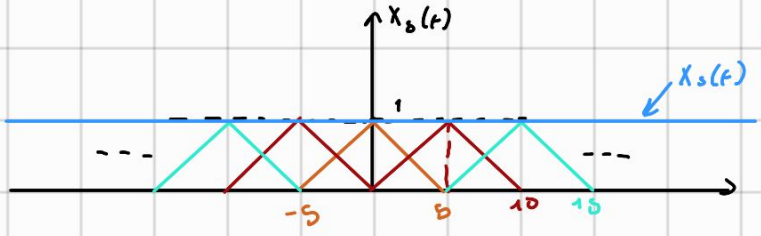
$k \in \mathbb{Z} - \{0\} \rightarrow t = \frac{k}{5}$ 'te sıfır

$$\text{sinc}\, k = \frac{\sin(\pi k)}{\pi t} = 0 \quad k \in \mathbb{Z} - \{0\}$$



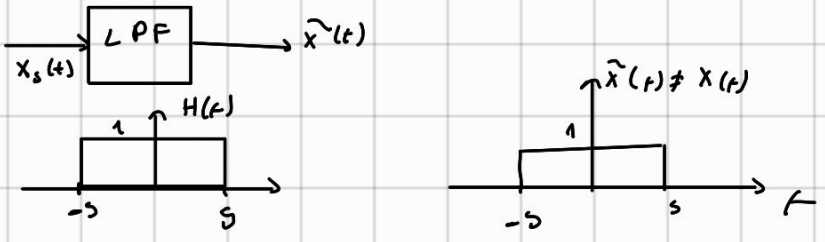


b)



c) No, $f_s = 5\ Hz < 2f_m = 10\ Hz$

d)

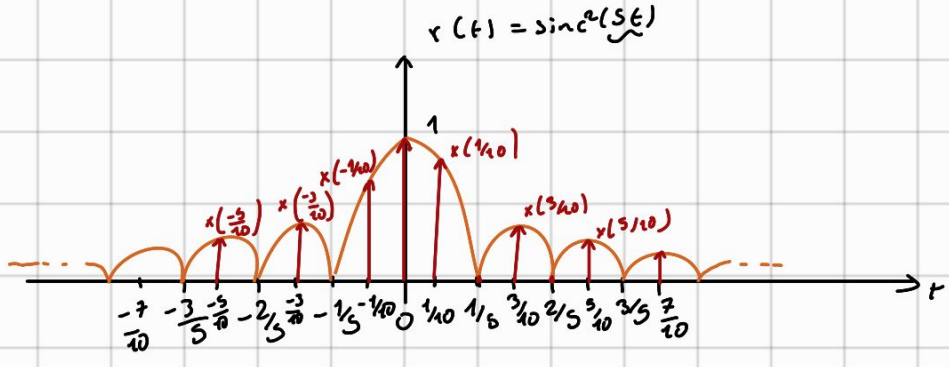


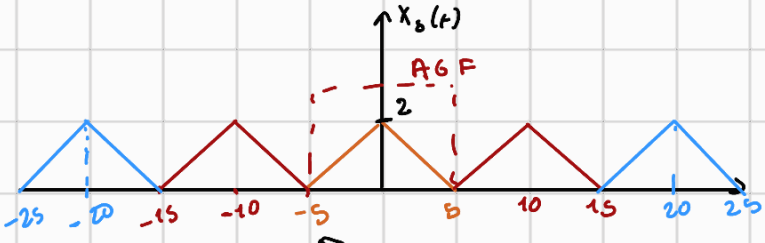
(ii) $f_s = 10\ Hz$, $T_s = \frac{1}{f_s} = \frac{1}{10}\ sn$

$$x_\delta(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \rightarrow 1/10$$

$$x_s(t) = x(t)\, x_\delta(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\, \delta(t - nT_s)$$

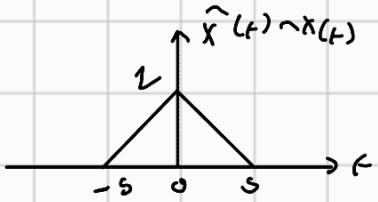
$$X_s(t) = \frac{f_s}{10} \sum_{n=-\infty}^{\infty} X(f - n f_s)$$

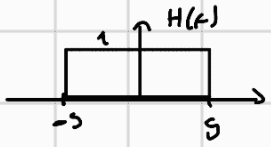




c) Yes $f_s = 10\ Hz = 2f_m = 10\ Hz$

d)





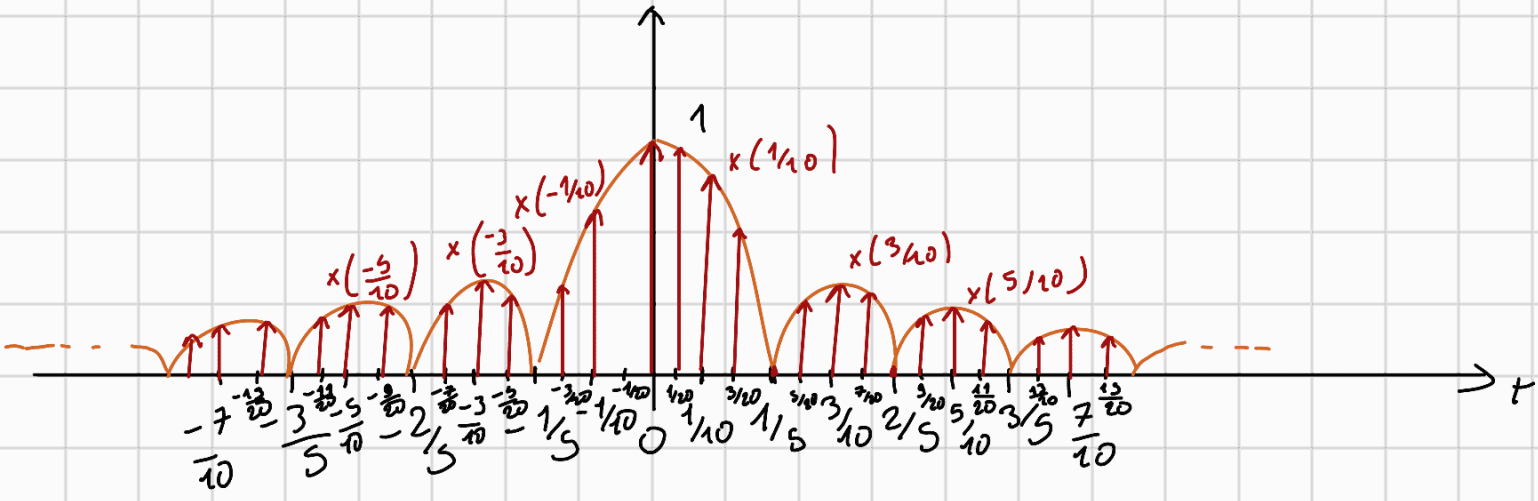
ii) $f_s = 20\ Hz$, $T_s = \frac{1}{f_s} = \frac{1}{20}\ sn$

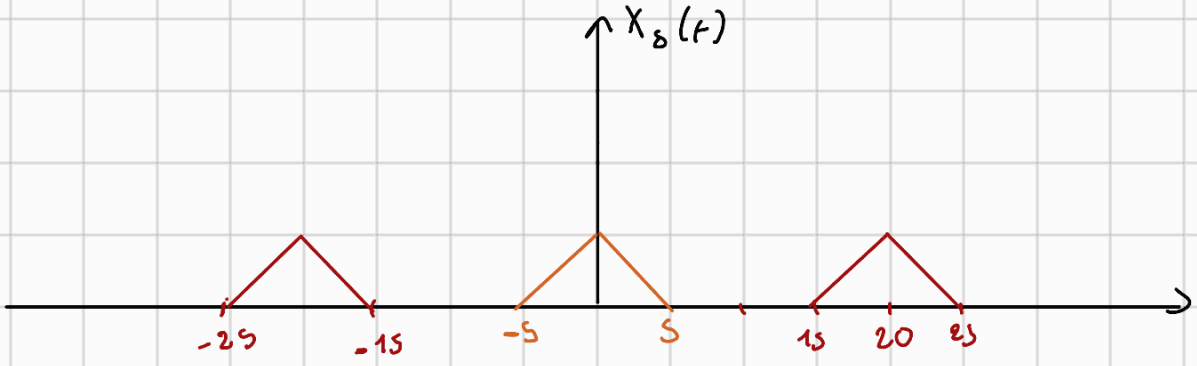
$$x_\delta(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \quad \rightarrow 1/20$$

$$x_s(t) = x(t)\, x_\delta(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\, \delta(t - nT_s)$$

$$X_s(f) = \frac{f_s}{20} \sum_{n=-\infty}^{\infty} X(f - n f_s)$$

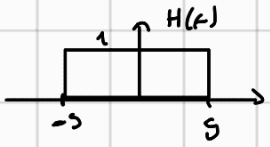
$x(t) = \mathrm{sinc}^2(5t)$

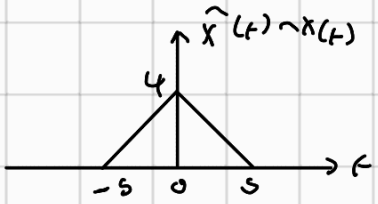




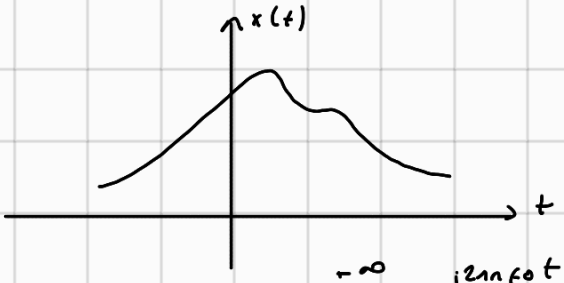
c) Yes $f_s = 20\,Hz > 2f_m = 10\,Hz$

d)

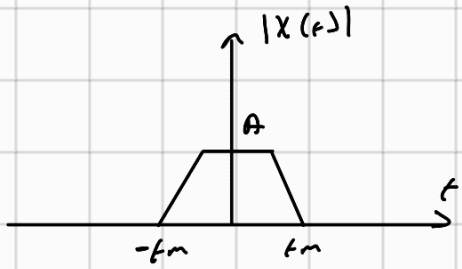




Pratik örnekleme
(Natural sampling)

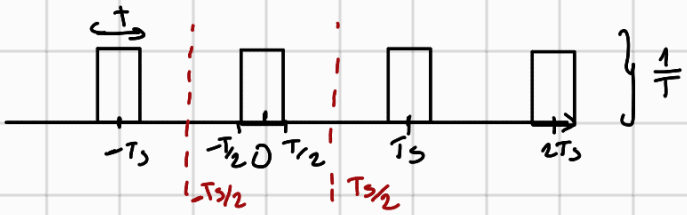


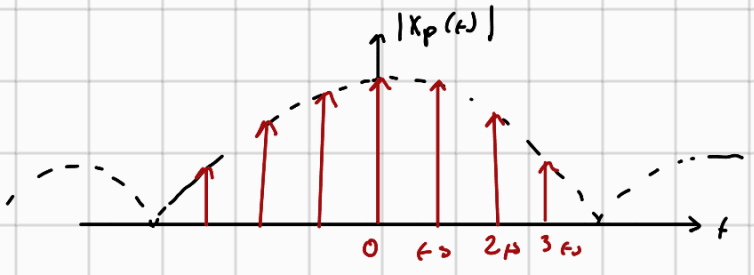
$\xleftrightarrow{F}$

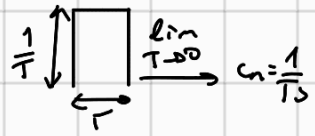


$$x_p(t) = \sum_{n=-\infty}^{+\infty} c_n e^{j2\pi n f_s t}$$

Darbe katarı (Pulse train)







$$x_p(t) = \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_s t}$$

$$c_n = \frac{1}{T_s}\int_{-T_s/2}^{T_s/2} x_p(t)\, e^{-j2\pi n f_s t}\, dt = \frac{1}{T_s}\int_{-T/2}^{T/2} \frac{1}{T}\, e^{-j2\pi n f_s t}\, dt$$

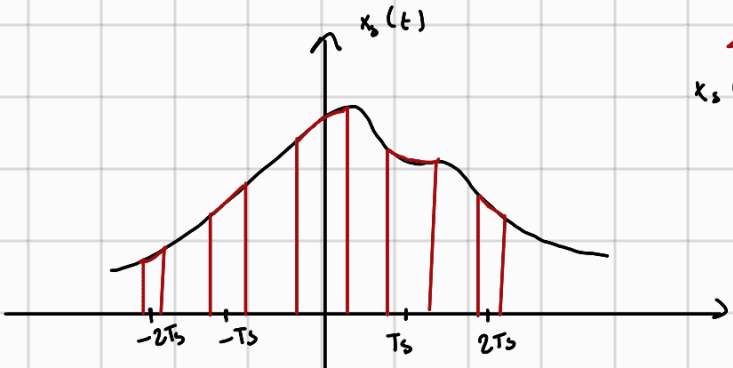
$$= \frac{1}{T_s}\cdot\frac{1}{T}\cdot\frac{e^{-j2\pi n f_s t}}{-j2\pi n f_s}\Bigg|_{-T/2}^{T/2} = \frac{1}{T_s}\cdot\frac{1}{T}\cdot\frac{e^{-j\pi n f_s T} - e^{j\pi n f_s T}}{-j2\pi n f_s}$$

$$= \frac{1}{T_s}\,\frac{\sin(\pi n f_s T)}{\pi n f_s T} = \frac{1}{T_s}\,\mathrm{sinc}(n f_s T)$$

$$c_n = \frac{1}{T_s}\,\mathrm{sinc}(n f_s T)$$

$$X_p(f) = \sum_{n=-\infty}^{\infty} c_n\, \delta(f - n f_s) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} \mathrm{sinc}(n f_s T)\,\delta(f - n f_s)$$

(DGM)
Darbe Genlik mod. (eniliger analogtejill! sonra)
Pulse amplitude mod (PAM)

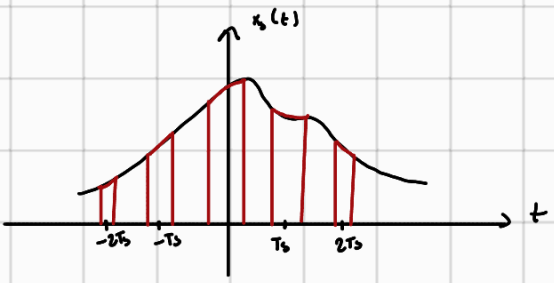


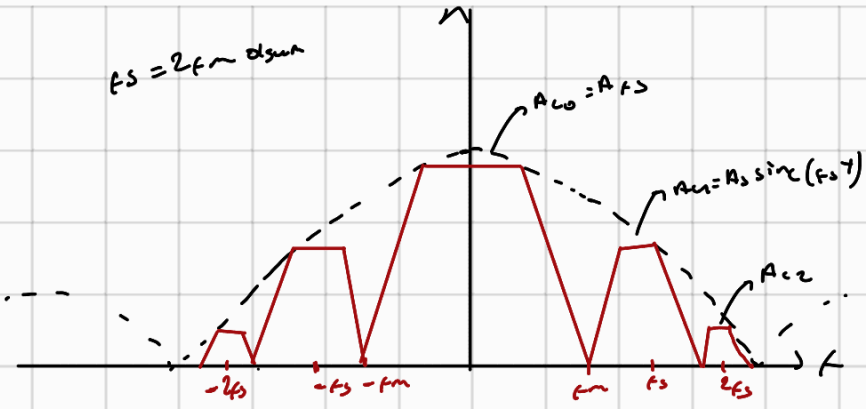
$$x_s(t) = x(t)\,x_p(t) = x(t)\sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_s t}$$

$$x_s(t) = \sum_{n=-\infty}^{\infty} c_n\, x(t)\, e^{j2\pi n f_s t}$$

F ↓ $\quad c_n = f_s \cdot \mathrm{sinc}(n f_s T)$

$$X_s(f) = \sum_{n=-\infty}^{\infty} c_n\, X(f - n f_s)$$





$$x_s(t) \longleftrightarrow X_s(f) = X(f) \circledast X_p(f) = X(f) \circledast \sum_{n=-\infty}^{\infty} c_n \delta(f - nf_s)$$

$$= \sum_{n=-\infty}^{+\infty} c_n X(f - nf_s)$$

Örnek Lathi, Pr: 6.1-3

A signal $x(t)$ band limited to B Hz is sampled by a periodic pulse train $p_{T_s}(t)$ made up of a rectangular pulse of width $\frac{1}{8B}$ seconds (centered at the origin) repeating at the Nyquist rate (2B pulses per second). Show that the sampled signal $x_s(t)$ is given by

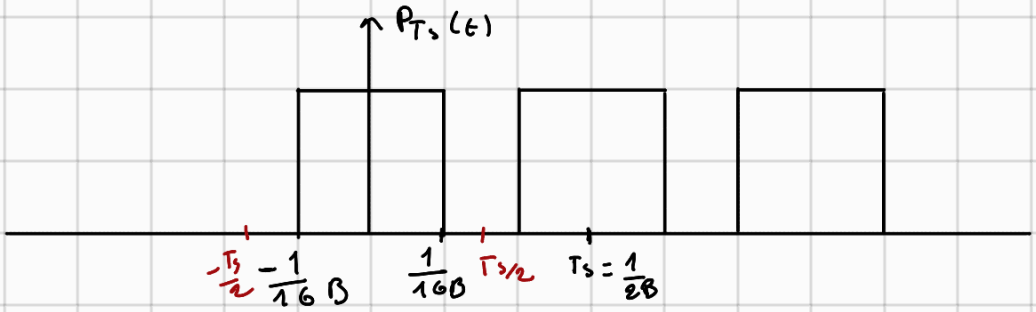
$$x_s(t) = \frac{1}{4} x(t) + \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin\left(\frac{n\pi}{4}\right) x(t) \cos(n w_s t), \quad w_s = 2\pi f_s = 4\pi B$$

Show that the signal $x(t)$ can be recovered by passing $x_s(t)$ through an ideal LPF of BW B Hz and a gain of 4.

D&P

$f_s = 2B$

$T_s = 1/2\ B$



$$x_s(t) = x(t) \cdot p_{T_s}(t)$$

$$p_{T_s}(t) = \frac{1}{4} + \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin\left(n\frac{\pi}{4}\right) \cos(n w_s t) \qquad w_s = 2\pi f_s = 4\pi B$$

böyle olduğu ispatlamamız gerek

$$p_{T_s}(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n w_s t) + b_n \sin(n w_s t)$$

$$a_0 = \frac{1}{T_s} \int_{T_s} x(t)\,dt = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} x(t)\,dt = 2B \int_{-1/16B}^{1/16B} 1\,dt = 2B \cdot \frac{2}{16B}$$

$$= 1/4$$

$b_n = 0$ çünkü consigonciyt (p_{T_s} çift $\rightarrow b_n = 0$)

$$a_n = \frac{2}{T_s}\int_{-T_s/2}^{T_s/2} p_{T_s}(t)\cos(n\omega_s t)\,dt = 4B\int_{-1/16B}^{1/16B} 1\cdot\cos(n\omega_s t)\,dt$$

$$= 4B\,\frac{\sin(n\omega_s t)}{n\omega_s}\Bigg|_{-\frac{1}{16B}}^{1/16B} = \frac{1}{n\pi}\left[\sin\left(n\,4\pi B\cdot\frac{1}{16B}\right) - \sin\left(n\,4\pi B\cdot\frac{-1}{16B}\right)\right]$$

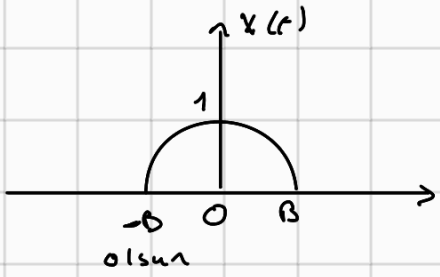
($\omega_s = 4\pi B$)

$$\boxed{a_n = \frac{2}{n\pi}\sin\left(n\frac{\pi}{4}\right)}$$

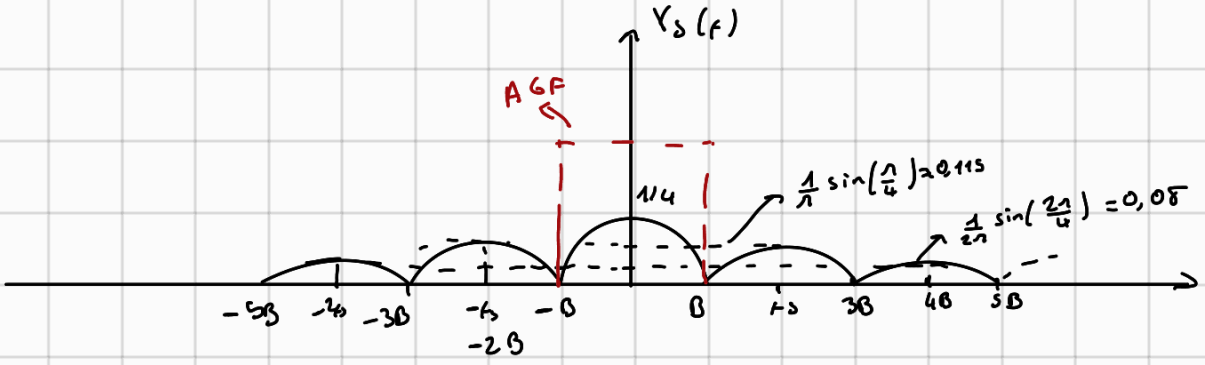
$$a_0 = \frac{\pi}{4} \rightarrow p_{T_s}(t) = \underbrace{\frac{1}{4}}_{a_0} + \underbrace{\sum_{n=1}^{\infty}\frac{2}{n\pi}\sin\left(\frac{n\pi}{4}\right)}_{a_n}\cos(n\omega_s t)$$

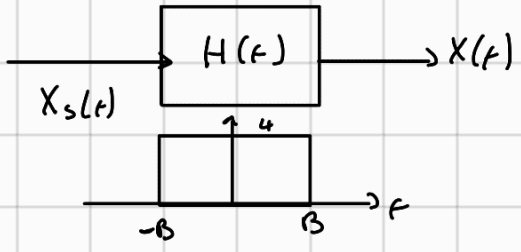
$$x_s(t) = g(t)\,p_{T_s}(t) = \frac{1}{4}x(t) + \sum_{n=1}^{\infty}\frac{2}{n\pi}\sin\left(\frac{n\pi}{4}\right)\underbrace{\cos(n\omega_s t)}_{e^{jnf_s t}}\,x(t)$$

b)



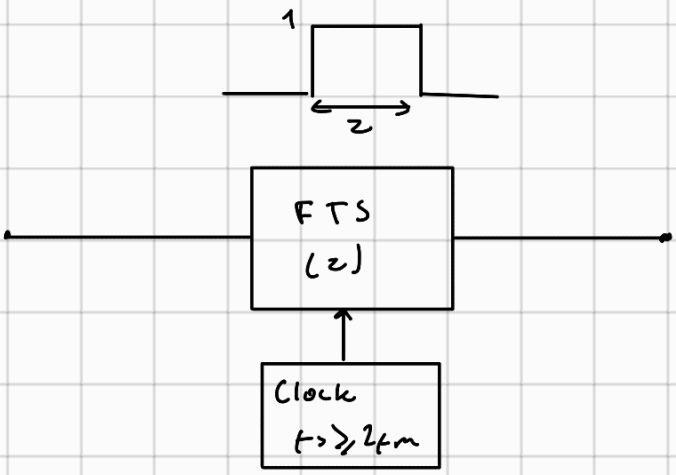
$$X_s(f) = \frac{1}{4}X(f) + \frac{1}{2}\frac{2}{n\pi}\sin\left(\frac{n\pi}{4}\right)\left[X(f-nf_s) + X(f+nf_s)\right]$$

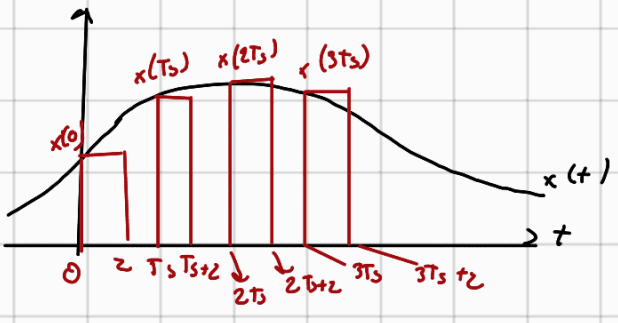




Düz Tepeli örnekleme
(Flat top sampling) (FTS)

tepeği takip etmiyor





Sample and hold circuit
(Örnekleve tut devresi)

$x_s(t)$: Darbe Genlikmod (DGM)

Pulse amplitudemod